

DEFINITION — HOW DATA IS DESCRIBED

CONTRACTS6

ADOPT



ODCS

BITOL @ LF

ADOPT



OpenAPI

LF

ADOPT



AsyncAPI

LF

ADOPT



GraphQL

LF

SITUATIONAL



gRPC

CNCF

SITUATIONAL



OData

OASIS

DATA PRODUCTS4

ADOPT



ODPS

BITOL @ LF

ASSESS



DPDS

ODM

ASSESS



ODPS

LF

ASSESS



DPROD

OMG

SCHEMA7

ADOPT



XML Schema

W3C

ADOPT



JSON Schema

IETF

ADOPT



SQL DDL

ISO/IEC

ADOPT



AVRO Schema

ASF

ADOPT



Protobuf

GOOGLE

ASSESS



LinkML

LINKML

ASSESS



Data Package Table Schema

FRICTIONLESS

SEMANTICS8

SITUATIONAL



RDF/OWL

W3C

SITUATIONAL



DCAT

W3C

SITUATIONAL



SKOS

W3C

SITUATIONAL



SHACL

W3C

SITUATIONAL



JSON-LD

W3C

ASSESS



OSI

OSI INITIATIVE

ASSESS



schema.org

W3C

ASSESS



ShEx

W3C

STORAGE — WHERE DATA LIVES

FILE FORMATS8

ADOPT



CSV

IETF

ADOPT



JSON

IETF

ADOPT



XML

W3C

ADOPT



YAML

YAML.ORG

ADOPT



PARQUET

ASF

ADOPT



AVRO

ASF

SITUATIONAL



ORC

ASF

ASSESS



Lance

LANCE

OPEN TABLE FORMATS4

ADOPT



Iceberg

ASF

SITUATIONAL



Delta

LF

ASSESS



Hudi

ASF

ASSESS



Lance

LANCE

STORAGE SYSTEMS2

ADOPT



S3

AWS

CAUTION



HDFS

ASF

MOVEMENT — HOW DATA FLOWS BETWEEN SYSTEMS

DATABASE CONNECTIVITY5

ADOPT



JDBC

JCP

ADOPT



ODBC

ISO/IEC

ADOPT



ADBC

ASF

ASSESS



Quack

DUCKDB LABS

CAUTION



XMLA

MICROSOFT

INTERCONNECTION3

ADOPT



HTTP

IETF

SITUATIONAL



Delta Sharing

LF

CAUTION



FTP / SFTP

IETF

MESSAGING5

ADOPT



Kafka

ASF

ADOPT



CloudEvent

s CNCF

SITUATIONAL



MQTT

OASIS

SITUATIONAL



AMQP

OASIS

CAUTION



JMS

JAKARTA EE

TRANSFORMATION — HOW DATA IS PROCESSED AND RESHAPED

IN-MEMORY FORMAT2

ADOPT



DataFrame API

DATA-APIs

ADOPT



Apache Arrow

ASF

PROCESSING7

ADOPT



Spark

ASF

ADOPT



Pandas

NUMFOCUS

ADOPT



SQL DML

ISO/IEC

ADOPT



dbt

DBT LABS

SITUATIONAL



Beam

ASF

ASSESS



Ibis

IBIS

CAUTION



XSLT

W3C

DISCOVERY — HOW DATA IS FOUND AND TRACED

CATALOG APIS5

ADOPT



Iceberg Catalog

ASF

ADOPT



Schema Registry

CONFLUENT

SITUATIONAL



Unity Catalog

LF

ASSESS



DuckLake

DUCKDB LABS

CAUTION



Hive Metastore

ASF

LINEAGE2

ADOPT



OpenLineage

LF

ASSESS



PROV

W3C

OPERATIONS — HOW DATA IS QUERIED, OBSERVED, GOVERNED

QUERY5

ADOPT



SQL

ISO/IEC

SITUATIONAL



Substrait

LF

SITUATIONAL



SPARQL

W3C

SITUATIONAL



GQL

ISO/IEC

CAUTION



MDX

MICROSOFT

DATA QUALITY3

SITUATIONAL



Great Expectations

GX LABS

SITUATIONAL



dbt tests

DBT LABS

SITUATIONAL



SodaCL

SODA

OBSERVABILITY2

ADOPT



OpenTelemetry

CNCF

ASSESS



OORS

BITOL @ LF

POLICIES2

ADOPT



OPA

CNCF

ASSESS



ODRL

W3C

AI INTERFACES2

ADOPT



MCP

LF

SITUATIONAL



A2A

LF